SEAN MACBRIDE

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EDUCATION

Ph.D. in Physics, University of Zürich — May 2024 - 2027

Thesis title: *Standard Siren Cosmology: From Instrumentation to Inference.*
Advisor: Professor Marcelle Soares-Santos.

M.Sc. in Physics, University of Michigan-Ann Arbor — Aug 2022 - May 2024

Advisor: Professor Marcelle Soares-Santos.

B.A. in Physics with Honors, Wheaton College, Massachusetts — Aug 2016 - May 2020

Thesis title: *Characterizing the Dust and Cold-Gas Content of Nearby Star-Forming Galaxies.*
Advisor: Professor Dipankar Maitra.

PROFESSIONAL APPOINTMENTS AND SERVICE

2026 - 2028	LSST DESC, Speakers Bureau Co-Chair.
2026 - 2028	LSST DESC, Collaboration Council Member.
2025 -	Rubin Observatory, Target-of-Opportunity (ToO) Observer.
2024 -	Rubin Observatory, LSST-Camera (LSSTCam) Summit Support Scientist.
2022 - 2024	University of Michigan Physics Graduate Council, Elected member.
2022 - 2024	University of Michigan Physics DEI committee, Member.
2022 - 2024	University of Michigan APS chapter, Member.

SELECTED PUBLICATIONS

Refereed Publications

1. Measurement of the Hubble constant using the Dark Energy Survey Year 6 Gold galaxy catalog and the fourth Gravitational-Wave Transient Catalog. McMahon, ... **MacBride**, ... 5 Feb 2026. Summary of contributions: Supported galaxy catalog preparation, including coincidence metric to determine the most significant gravitational wave (GW) events in the DES footprint.
2. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1). Chandler, Bernardelli, ... **MacBride**, ... 21 July 2025. Summary of contributions: Led ToO observations of 3I/ATLAS to obtain full grizy color profile and obtain constraints on orbital period.

Preprints

1. Constraints on the Cosmic Expansion Rate Using the DECam Local Volume Exploration Survey and the Fifth Gravitational Wave Transient Catalog. **MacBride**, McMahon, ... 19 Oct. 2026, submitted to Astronomy & Astrophysics.
2. A Joint Search for the Electromagnetic Counterpart to the Gravitational-Wave Binary Black-Hole Merger Candidate S250328ae with the Dark Energy Camera and the Prime Focus Spectrograph. Zhang, Kokubo, **MacBride**, ... 1 Aug. 2025, submitted to ApJ. Summary of contributions: Led ToO observations using DECam, coordinated PFS observations, analyzed obtained spectroscopic & photometric data to identify transients.

Technical Proceedings

1. The On-Sky Performance of the LSST Camera CCD Array. **MacBride**, Roodman, ... 30 Jun 2026.
2. The Rubin Observatory Target-of-Opportunity Program in the First Year of Operations. **MacBride**, Ribeiro, ... 30 Jun 2026.
3. First year into operations of LLAMAS, the Large Lenslet Array Magellan Spectrograph: design, build and commissioning overview, results and performance, lessons learned. Furesz, ... **MacBride**, ... 21 Aug 2026.
4. Characterizing robotic positioners under the influence of changing gravity vectors for future spectroscopic surveys. Wüthrich, ... **MacBride**, ... 25 Jun 2026.
5. Scheduling the LSST: commissioning the Rubin Feature Based Scheduler. Jones, ... **MacBride**, ... 19 Aug 2026.

INVITED TALKS

07/2026	Plenary, Rubin Community Workshop 2026, SLAC National Accelerator Laboratory, Menlo Park, CA, USA.
07/2026	Seminar, Dark Energy School, LSST DESC Collaboration Meeting, Paris, France.
07/2026	Talk, SPIE Astronomical Telescopes + Instrumentation, Copenhagen, Denmark.
11/2025	Talk, Rubin-LSST France Meeting at LPSC Grenoble, France.
09/2025	Talk, 5th Philip Wetton Workshop, Oxford University, Oxford, Great Britain.

APPROVED OBSERVING PROGRAMS

2025	Gravitational Wave Photometric Redshift Improvement with Medium-band Observations (GW-PRIMO). PI: McMahon, Co-I: MacBride. 2 nights on the Blanco telescope at CTIO.
2023 - 2027	Dark Energy Survey Gravitational Wave Follow-up Program (DESGW). PI: Soares-Santos, Co-I: MacBride. 3.2 nights on the Blanco telescope at CTIO, every 6 months from 2023B - 2027A.

RESEARCH GRANTS

2024 - 2026	*Quantifying the Impact of Galaxy Catalog Conditions for Dark Sirens*, 3M CPU hour grant National Energy Research Scientific Computing Center, ScienceAtScale.

TEACHING

2026	Introduction to Astroparticle Physics (PHY473), University of Zürich, Graduate Instructor.
2026	Introduction to Scientific Computing (PHY124), University of Zürich, Graduate Instructor.
2025	Introduction to Astroparticle Physics (PHY473), University of Zürich, Graduate Instructor.
2023 - 2024	Introductory Physics for Life Sciences (PHYSICS 151/251), University of Michigan-Ann Arbor, Lead Graduate Instructor.
2022 - 2023	Introductory Physics for Life Sciences (PHYSICS 151/251), University of Michigan-Ann Arbor, Graduate Instructor.

MENTORSHIP

2026	Alexs Gaute: Summer Masters Student at University of Zürich.
2025 - 2026	Guandi Zhao: Masters Student at University of Zürich. Current Physics PhD student at University of Zürich.
2024 - 2025	Banan Yamani: Masters Student at University of Zürich.
2024 - 2025	Yavuz Gencel: Masters Student at University of Zürich.
2023 - 2024	Elise Fuller: Undergraduate student at University of Michigan-Ann Arbor. Current physics PhD student at CU Boulder.
2023 - 2024	Tim Fanning: Undergraduate student at University of Michigan-Ann Arbor. Current controls engineer specialist at Argonne National Laboratory.

OUTREACH

2026	Led 'Ask a Scientist' sessions focused on observational cosmology and gravitational waves for the Science Pavilion at UZH, as part of the City of Zurich's Long Night of Museums, an annual event where Zurich's museums invite visitors to engage with science, culture, and art museums free of charge.
2025	Led tour groups and interviews with media members at Rubin Observatory, focusing on LSSTCam. Visting parties included including media members, friends and family visits, tour groups, visiting project review teams from the National Science Foundation and Department of Energy, and political dignitaries.
2023 - 2024	Led group of high school students in Burlington VT to observe the 2023 annular eclipse and 2024 total eclipse. Prepared and delivered lectures for students about the physics of the sun, solar observations, and eclipses through virtual and in person presentations.
2023	Designed a display in the University of Michigan Physics help room, highlighting struggles overcame by recent graduates of the physics program and success in their later lives.
2016 - 2020	Showcased features of different telescopes to local tour groups at Wheaton College Observatory. Operate and adjust telescopes to show appropriate stars, objects, and events based on sky conditions. Educated children and adults in how to locate objects and explained features about the objects on sky.

CONFERENCE ORGANIZATION

2026	EAS Annual Meeting Special Session 27: The Rubin Observatory Legacy Survey of Space and Time: from Data Previews to Full Survey Operations, co-chair, local organizing committee.
2024	Conferences for Undergraduate Women in Physics (CUWiP) 2024, local organizing committee.
2023	DESI Collaboration Meeting, local organizing committee.
2023	University of Michigan-Ann Arbor Physics Graduate Student Symposium, chair local organizing committee.